# AT 2020afkm: a dust-obscured infrared transient in NGC 3239 found in archival WISE data, with a fading Hubble counterpart and a JWST dust remnant

Alan Shortz (Oddsky) · TNS AstroNote · Related object: AT 2020afkm (RA 10:25:09.894, Dec +17:09:49.60) · November 2026

**Abstract.** We report AT 2020afkm, a dust-obscured mid-infrared transient in NGC 3239 (about 7-12 Mpc), found in a blind difference-imaging search of the unWISE time-resolved coadds of 318 nearby galaxies. Quiet from 2010 to 2020-04-28, it was detected on 2020-11-23 at W1 = 17.99 and W2 = 17.67 AB (about 960 K dust; about 5.6e5 L_sun at 7.9 Mpc) and faded over the following year. Public Hubble images from 2021-02-16 show a point source at the position (F814W = 24.95 AB) that is gone by 2022, and JWST/NIRCam images from 2024 show a new, cooler (about 580 K) dust source there. No optical, radio or X-ray counterpart is known. A normal supernova hidden by dust fits poorly; an obscured luminous red nova, an ILRT-like eruption or a low-energy supernova behind clumpy dust remain possible. JWST spectroscopy of the remnant could decide.

We report the discovery of AT 2020afkm, a mid-infrared transient in the irregular galaxy NGC 3239 (Arp 263; z = 0.00237, distance about 7-12 Mpc), found in a blind difference-imaging search of 318 star-forming galaxies within 10 Mpc. The search used the unWISE time-resolved coadds (NEO11 version, courtesy A. Meisner; Meisner, Lang & Schlegel 2018), 25 epochs from 2010 to 2024. As an acceptance test, the same search recovers the failed-supernova candidates M31-2014-DS1 (De et al. 2024) and N6946-BH1 blind. No optical transient was reported at this position.

**WISE.** Nothing is seen at the position in the 15 epochs from 2010 to 2020-04-28 (last non-detection 2020-04-28: W1 > 19.9 AB, 5 sigma). On 2020-11-23 the source is detected at W1 = 17.99 +- 0.04 and W2 = 17.67 +- 0.07 AB (difference fluxes 232 +- 8 and 310 +- 20 uJy; 28 and 15 sigma), and then fades: W1 = 18.96 and W2 = 18.63 AB on 2021-04-30, W1 = 19.92 AB on 2021-11-22, near the detection limit by 2022. The field was in solar conjunction for much of the gap, so the rise (< 209 days) and the true peak are not constrained; the first detection is a lower limit on the peak. The colour W1 - W2 = 0.95 +- 0.08 (Vega) at the first two epochs corresponds to a blackbody of about 960 K, i.e. warm dust. The blackbody luminosity at the first detection is about 5.6e5 L_sun at 7.9 Mpc (about twice that at 11 Mpc), and at least 3e46 erg was radiated in the infrared afterwards. The W1 decline e-folds in about 170 days and slows later.

**Hubble.** Public WFC3/UVIS images taken on 2021-02-16 (programme 16179, PI Filippenko; 85 days after the first WISE detection) show a point source at RA, Dec = 10:25:09.894, +17:09:49.60 (+- 0.004" statistical, Gaia-tied), 0.37" from the WISE centroid (+- 0.2-0.3"). In difference images it has F606W = 25.93 +- 0.06 and F814W = 24.95 +- 0.06 AB (M_F814W about -4.5 at 7.9 Mpc) and is detected in each of the four exposures. It is absent from ACS F606W images of 2022-12-11 (programme 15446, PI Dalcanton; > 27.1 AB) and from WFC3 images of 2025-01-09 (programme 17502, PI Thilker; F555W > 26.6 AB), with no residual star or cluster (M_V > -2.9). The chance that an unrelated source that vanishes like this lies within 0.37" is about 0.2%.

**JWST.** Public NIRCam images from 2024-04-24 and 2024-05-22 (programme 3707, PI Leroy) show an unresolved source 0.01" from the Hubble position at F300M = 22.65/22.50 and F335M = 21.97 AB, with nothing in F150W (> 26.5 AB). A blackbody fit gives about 580 K and 3.6e4 L_sun: a cooler, fainter dust source that was not there before the event. Spitzer/IRAC in 2008 shows nothing at the position (nu L_nu(4.5 um) < 4e3 L_sun, 3 sigma), which also excludes an infrared-bright progenitor like that of SN 2008S.

**Other limits.** ATLAS has no detection (3 sigma limits o > 20.4-21.7 in 2020 Oct to 2021 Jan); VLASS gives < 0.36 mJy at 3 GHz on 2021-11-20; Chandra (2025) gives L_X < 3e37 erg/s. The source is 70.6" (about 2.7 kpc) from the centre of a galaxy with no active nucleus.

**Interpretation.** The event is brighter in the infrared than the SPIRITS "SPRITEs" (M_[4.5] -11 to -14; Kasliwal et al. 2017), but a normal core-collapse supernova fully hidden by dust does not fit well: the Hubble colour (V - I about 1.5) allows only A_V of about 1-2, while the faintness relative to the infrared would need A_V of about 9-20, and a fully enshrouded Type II supernova should re-radiate 30-100 times more infrared light at this phase. A dust-obscured luminous red nova (stellar merger), an ILRT-like eruption without a bright progenitor, or a low-energy supernova behind clumpy dust all remain possible. The JWST remnant (F335M = 22.0 AB) is bright enough for NIRSpec and MIRI spectroscopy, which could distinguish these: molecular bands such as CO and H2O would point to a merger, and [Ni II]/[Fe II] lines or silicate dust to a supernova. We encourage such follow-up.

Oddsky (oddsky.ai) is an open project that searches public sky-survey archives for unreported astronomical events; the full analysis, data and code for this object are public as Oddsky case OS-0030.

**Acknowledgements:** unWISE time-resolved coadds (NEO11 version, courtesy A. Meisner; Meisner, Lang & Schlegel 2018). This work uses data from WISE/NEOWISE (NASA/JPL/IPAC, via IRSA), from the NASA/ESA Hubble and NASA/ESA/CSA James Webb Space Telescopes obtained from MAST (programmes HST 16179, 15446, 17502 and JWST 3707), from Spitzer (IRSA), ATLAS, VLASS (NRAO, via CADC) and Chandra.

## Figures

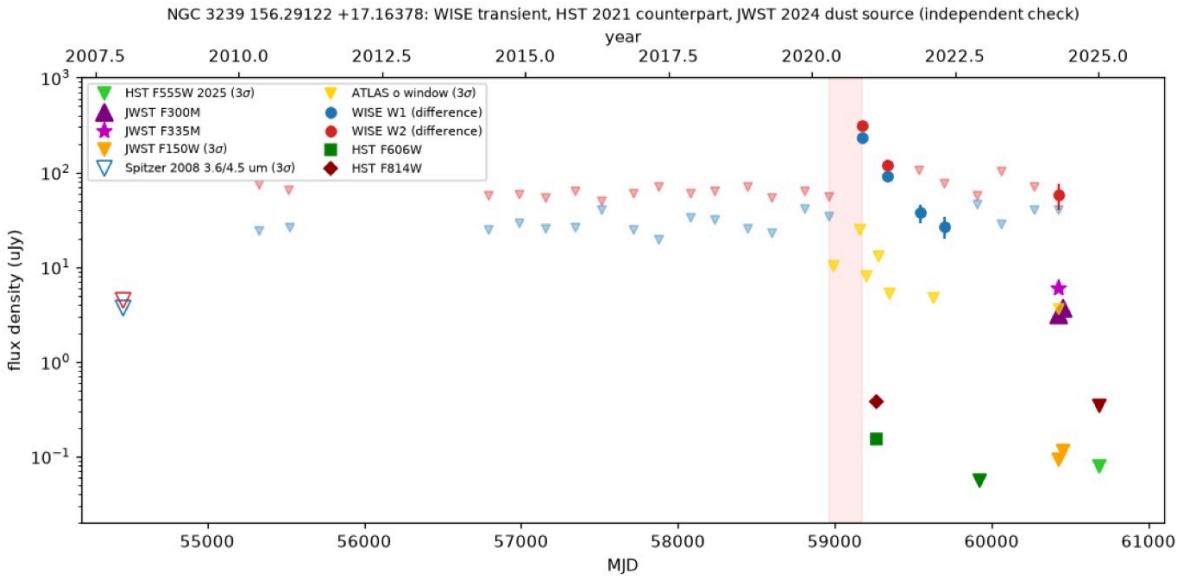



Figure 1. Overview of AT 2020afkm.

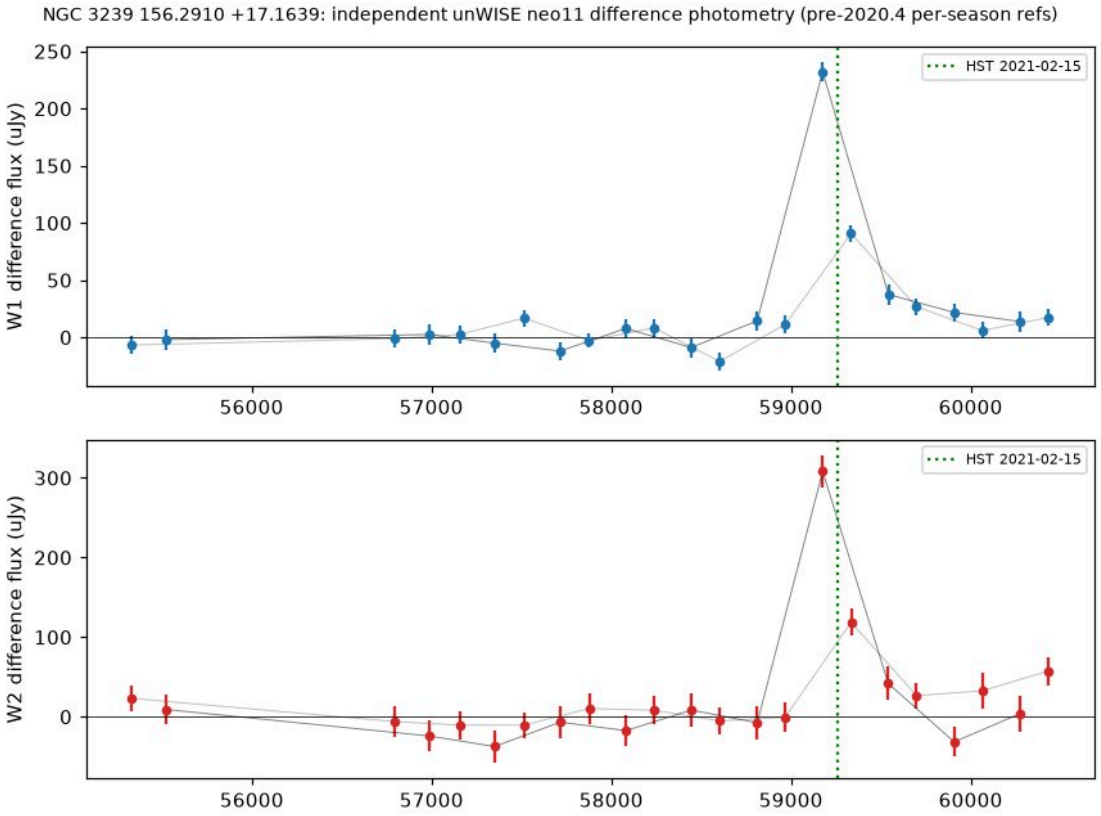



Figure 2. unWISE W1 (top) and W2 (bottom) difference light curves at the position, 2010-2024 (x-axis: MJD; dotted line: the Hubble epoch).

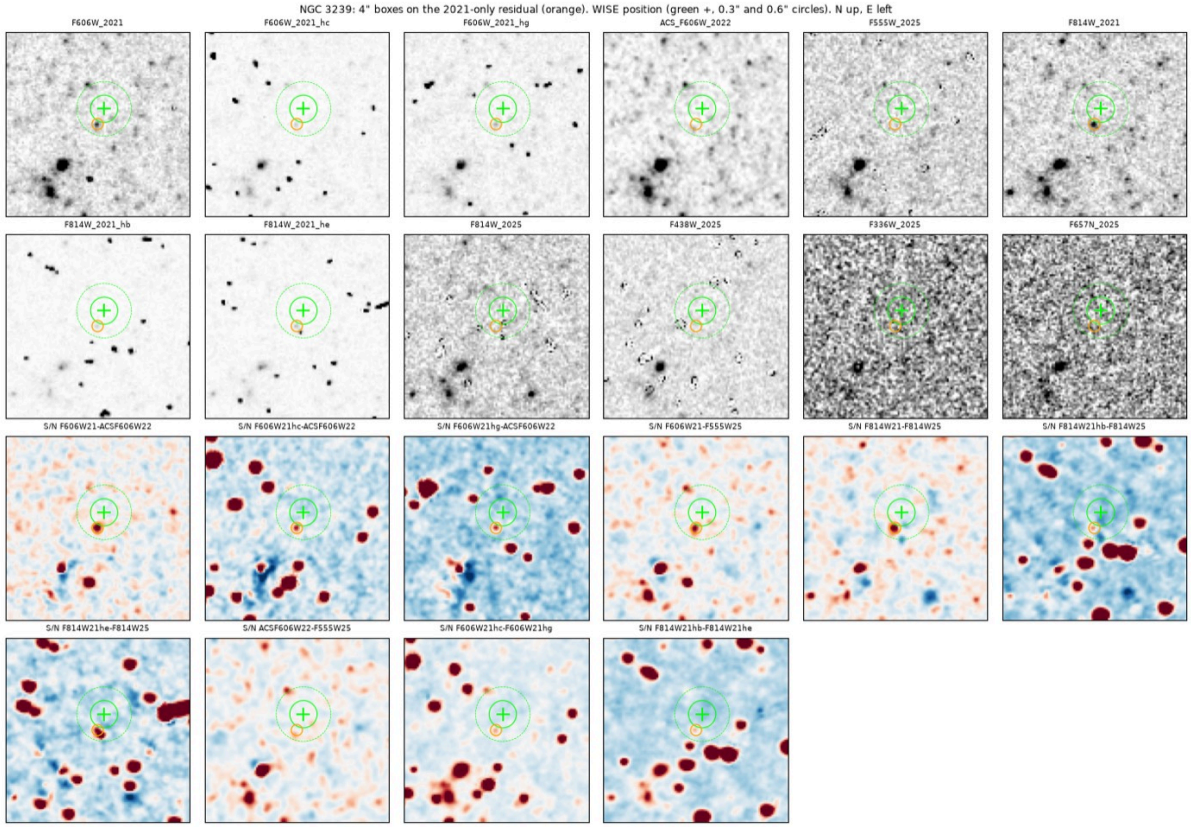



Figure 3. Hubble at the position: 2021-02-16 (point source), 2022-12-11 and 2025-01-09 (absent).

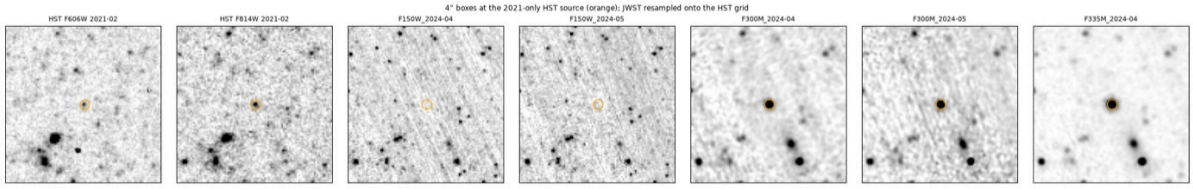



Figure 4. JWST/NIRCam 2024: F150W (nothing), F300M and F335M (new warm-dust source).

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